**PEM 2020 Year 11 Mathematics Standard Yearly Examination**

**SUGGESTED SOLUTIONS**

## Section 1 Multiple Choice Answer Key

| Question | Answer |
|---|---|
| 1 | *C* |
| 2 | *A* |
| 3 | *C* |
| 4 | *D* |
| 5 | *B* |
| 6 | *B* |
| 7 | *D* |
| 8 | *C* |
| 9 | *B* |
| 10 | *C* |
| 11 | *B* |
| 12 | *B* |

## **Section II** (Total 68 marks)

### Question 13

| Criteria | Marks |
|---|---|
| Provides correct solution | 2 |
| Makes progress towards the solution | 1 |

*Sample Answer:*

$$c = \sqrt{6^2 + 8^2}$$
$$= 10$$

### Question 14 (a)

| Criteria | Marks |
|---|---|
| Provides correct solution | 2 |
| Makes progress towards the solution by calculating deposit or repayments. | 1 |

*Sample Answer:*

Deposit $= 11999 \times 25\% = \$2999.75$.

Repayments $= 405 \times 12 \times 3 = \$14580$

Total $= \$2999.75 + \$14580 = \$17579.75$

### Question 14 (b)

| Criteria | Marks |
|---|---|
| Provides correct solution | 1 |

*Sample Answer:*

$17579.75 - 11999 = \$5580.75$. Julie is charged $\$5580.75$ in interest.

**Question 15 (a)**

| Criteria | Marks |
|---|---|
| Provides correct solution | 1 |

*Sample Answer:*

$3.3\times 2+2.5\times 2+4.2\times 2=20m$.

$20\times \$15=\$300$

**Question 15(b)**

| Criteria | Marks |
|---|---|
| Provides correct solution | 1 |

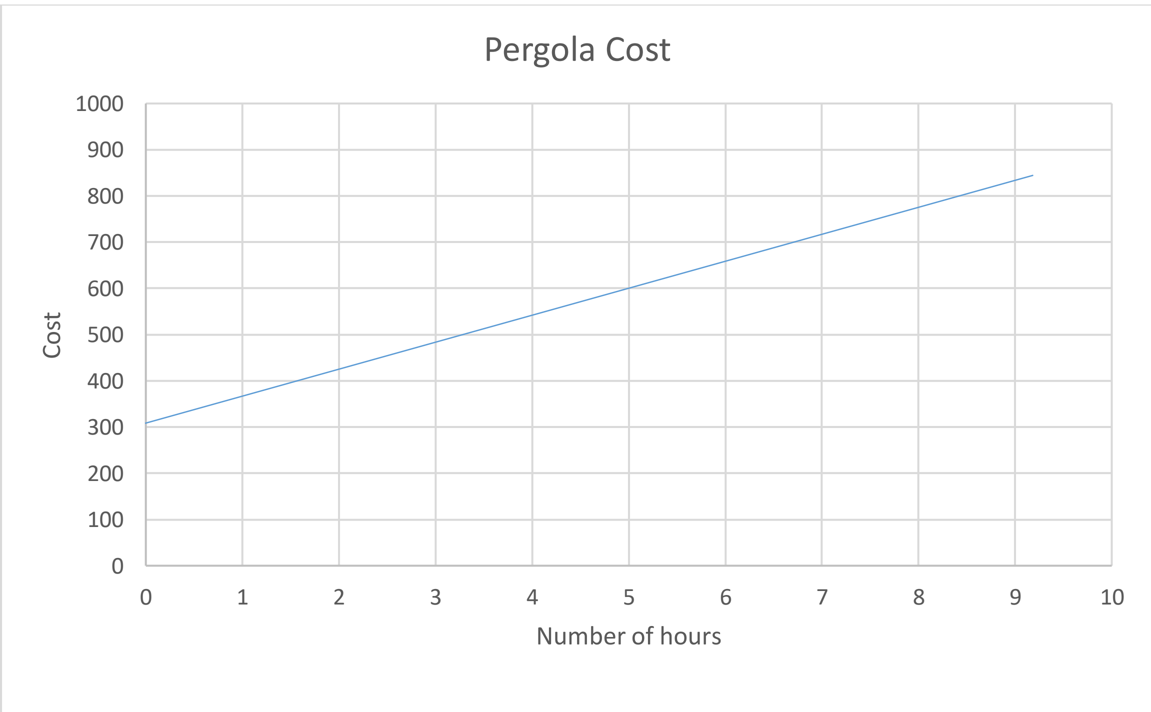
*Sample Answer:*

$300+60\times 2=\$420$

**Question 15 (c)**

| Criteria | Marks |
|---|---|
| Provides correct solution | 1 |

*Sample Answer:*



**Question 15 (d)**

| Criteria | Marks |
|---|---|
| Provides correct solution | 1 |

*Sample Answer:*

Approximately $510

**Question 16 (a)**

| Criteria | Marks |
|---|---|
| Provides correct solution | 2 |
| Makes progress towards the solution (1 correct) | 1 |

*Sample Answer:*

| Blood Type | Frequency | Relative Frequency |
|---|---|---|
| AB | 30 | 0.03 |
| A | 380 | 0.38 |
| B | 100 | 0.10 |
| $O^-$ | 90 | 0.09 |
| $O^+$ | 400 | 0.4 |

**Question 16 (b)**

| Criteria | Marks |
|---|---|
| Provides correct solution | 1 |

*Sample Answer:*

$0.09 = 9\%$

**Question 17**

| Criteria | Marks |
|---|---|
| Provides correct solution | 2 |
| Makes progress towards the solution | 1 |

*Sample Answer:*

New York is 5 – (-1) = 6 hours behind Sydney

Monday 6am + 15 hours is 9pm Monday in Sydney

Monday 8am + 15 hours is 11pm Monday in Sydney

She should ring between 9pm and 11pm on Monday

**Question 18(a)**

| Criteria | Marks |
|---|---|
| Provides correct solution | 2 |
| Makes progress towards the solution | 1 |

*Sample Answer:*

| Average Monthly Phone Cost | Class centre $(x)$ | Frequency $(f)$ | $fx$ |
|---|---|---|---|
| 1 - 50 | 25.5 | 6 | 153 |
| 51 - 100 | 75.5 | 7 | 528.5 |
| 101 - 150 | 125.5 | 13 | 1631.5 |
| 151 - 200 | 175.5 | 11 | 1930.5 |
| 201 - 250 | 225.5 | 18 | 4059 |
| 251 - 300 | 275.5 | 5 | 1377.5 |
| | | $\sum f = 50$ | $\sum fx = 9680$ |

**Question 18 (b)**

| Criteria | Marks |
|---|---|
| Provides correct solution | 1 |

*Sample Answer:*

9680

**Question 18 (c)**

| Criteria | Marks |
|---|---|
| Provides correct solution | 1 |

*Sample Answer:*

$9680 \div 50 = 193.6$

**Question 18 (d)**

| Criteria | Marks |
|---|---|
| Provides correct conclusion with reason | 1 |

*Sample Answer:*

Using the class centre instead of the actual scores to find the mean gives an approximation

**Question 19(a)**

| Criteria | Marks |
|---|---|
| Provides correct solution | 2 |
| Makes progress towards the solution by finding absolute error | 1 |

*Sample Answer:*

Absolute error $= \frac{1}{2} \times 0.1 = 0.05cm$

% Error = $\frac{0.05}{656.2} \times 100 = 0.0076\%$

**Question 19 (b)**

| Criteria | Marks |
|---|---|
| Provides correct solution | 2 |
| Makes some progress towards the solution by showing use of Pythagoras theorem | 1 |

*Sample Answer:*

$$\begin{aligned} s^2 &= 400^2 + 328.1^2 \\ &= \sqrt{400^2 + 328.1^2} \\ &= 517.3cm \end{aligned}$$

**Question 20 (a)**

| Criteria | Marks |
|---|---|
| Provides correct solution | 1 |

*Sample Answer:*

$$50+75=125$$

$$\frac{50}{125}=\frac{2}{5}=40\%$$

**Question 20 (b)**

| Criteria | Marks |
|---|---|
| Provides correct solution | 2 |
| Makes progress towards the solution by finding probability or showing use of expected value formula | 1 |

*Sample Answer:*

i) $P(GW)=\frac{3}{5}\times\frac{2}{5}=\frac{6}{25}=0.24$

ii) $n\times P(GW)=25\times\frac{3}{5}\times\frac{2}{5}=6$

**Question 21**

| Criteria | Marks |
|---|---|
| Provides correct solution | 2 |
| Makes progress towards the solution | 1 |

General Usage $71.50 \times 90 + 2080 \times 25.30 = 59059\, cents$

Solar Feed-in $390 \times 10.5 = 4095\, cents$

Bill $59059 - 4095 = 54964\, cents = \$549.64$

**Question 22**

| Criteria | Marks |
|---|---|
| Provides correct solution | 2 |
| Makes some progress towards the solution | 1 |

*Sample Answer:*

$$A_1 = \frac{h}{2}(d_f + d_l) = \frac{400}{2}(700 + 900) = 320000$$
$$A_2 = \frac{h}{2}(d_f + d_l) = \frac{400}{2}(900 + 650) = 310000$$
$$\text{Total} = 630000m^2$$

**Question 23 (a)**

| Criteria | Marks |
|---|---|
| Provides correct solution | 1 |

*Sample Answer:*

72%

**Question 23 (b)**

| Criteria | Marks |
|---|---|
| Provides correct solution | 1 |

*Sample Answer:*

Service

**Question 24 (a)**

| Criteria | Marks |
|---|---|
| Provides correct solution | 2 |
| Makes some progress towards the solution | 1 |

*Sample Answer:*

4 Schooners = 3.6 standard drinks

$$BAC_{male} = \frac{10\times3.6-7.5\times3}{6.8\times81} = 0.0245 = 0.025 \text{ to 2 significant figures}$$

**Question 24 (b)**

| Criteria | Marks |
|---|---|
| Provides correct solution | 1 |

*Sample Answer:*

$$T = \frac{0.0245}{0.015} = 1.63 \text{ hours} = 1 \text{ hour } 38 \text{ mins}$$

**Question 24 (c)**

| Criteria | Marks |
|---|---|
| Provides correct solution | 2 |
| Makes some progress towards the solution | 1 |

*Sample Answer:*

$y = mx + b \qquad b = 0$

$N = mS + b \qquad m = \dfrac{3.6-2.7}{1} = 0.9$

$\therefore N = 0.9S$

**Question 25(a)**

| Criteria | Marks |
|---|---|
| Provides correct solution | 2 |
| Makes progress towards the solution | 1 |

*Sample Answer:*

| Day | In | Out | Break | Number of Hours worked |
|---|---|---|---|---|
| Monday | Cafe closed | | | 0 |
| Tuesday | 9am | 3pm | 1 hour | 5 hours |
| Wednesday | 10am | 2pm | Half hour | 3.5 |
| Thursday | - | - | - | 0 |
| Friday | 7am | 10am | No break | 3 |
| Saturday | 7:30am | 2:30pm | Half hour | 6.5 |
| Sunday | 8:30am | 2pm | Half hour | 5 |

**Question 25(b)**

| Criteria | Marks |
|---|---|
| Provides correct solution | 2 |
| Makes progress towards the solution | 1 |

*Sample Answer:*

$$\text{Normal} = (5+3.5+3)\times 22 = \$253$$
$$\text{Saturday} = 6.5\times 1.5\times 22 = \$214.50$$
$$\text{Sunday} = 5\times 2\times 22 = \$220$$
$$\text{Total} = \$687.50$$

**Question 26 (a)**

| Criteria | Marks |
|---|---|
| Provides correct solution | 2 |
| Makes some progress towards the solution | 1 |

*Sample Answer:*

| | Minimum | Lower Quartile | Median | Upper Quartile | Maximum |
|---|---|---|---|---|---|
| Home | 12 | 20 | 45 | 51.5 | 64 |
| Away | 27 | 29.5 | 32 | 50 | 56 |

**Question 26 (b)**

| Criteria | Marks |
|---|---|
| Provides correct solution | 2 |
| Makes some progress towards the solution | 1 |

*Sample Answer:*

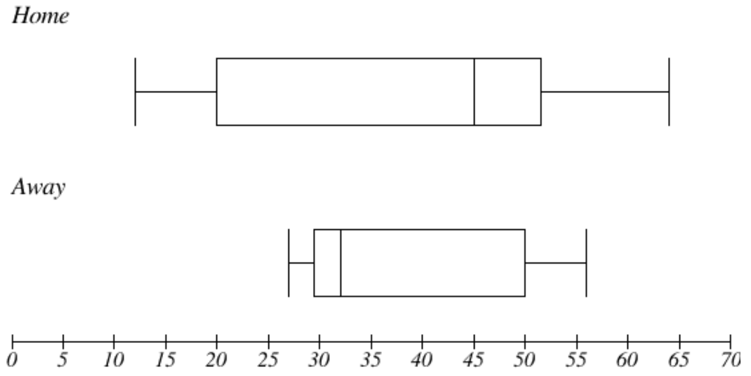
HOME: Range = 64-12 = 52, IQR = 51.5 – 20 = 31.5

AWAY Range = 56 – 27 = 29, IQR = 50 – 29.5 = 20.5

**Question 26 (c)**

| Criteria | Marks |
|---|---|
| Provides correct solution | 3 |
| Makes some progress towards the solution | 2 |
| Provides some relevant information | 1 |

*Sample Answer:*




**Question 26 (d)**

| Criteria | Marks |
|---|---|
| Provides correct solution | 2 |
| Makes progress towards solution by giving one comparison | 1 |

*Sample Answer:*

Similarities:

- Neither has any outliers

Differences:

- Away game attendance is positively skewed but Home game attendance is negatively skewed. Attendance for home games is higher than for Away games
- $IQR_{Home} = 31.5 > IQR_{Away} = 20.5$, i.e. The Attendance at Home games has the greater spread. Attendance at Home games is more consistent

**Question 27 (a)**

| Criteria | Marks |
|---|---|
| Provides correct solution | 1 |

*Sample Answer:*

$$S = \frac{D}{T} = \frac{30}{0.5} = 60km/h$$

**Question 27 (b)**

| Criteria | Marks |
|---|---|
| Provides correct solution | 1 |

*Sample Answer:*

$$d = \frac{5 \times 60 \times 0.75}{18} + \frac{60^2}{170} \quad = 33.68m$$

**Question 27 (c)**

| Criteria | Marks |
|---|---|
| Provides correct solution | 1 |

*Sample Answer:*

Yes

**Question 28 (a)**

| Criteria | Marks |
|---|---|
| Provides correct solution | 1 |

*Sample Answer:*

$C = 72 + 3 \times k$

**Question 28 (b)**

| Criteria | Marks |
|---|---|
| Provides correct solution | 1 |

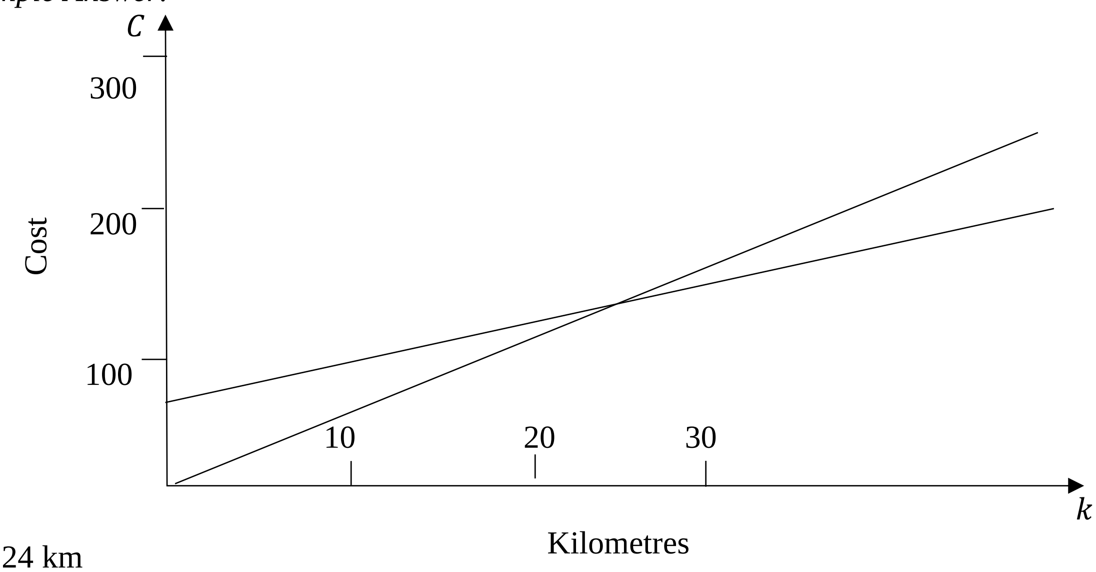
*Sample Answer:*

$C = 6 \times k$

**Question 28 (c)**

| Criteria | Marks |
|---|---|
| Provides correct solution | 3 |
| Makes some progress towards the solution | 2 |
| Provides some relevant information | 1 |

*Sample Answer:*




24 km

**Question 28 (d)**

| Criteria | Marks |
|---|---|
| Provides correct solution | 1 |

*Sample Answer:*

$k = 50 \times 5 \times 48 \times 0.68 = \$8160$ per year

**Question 29 (a)**

| Criteria | Marks |
|---|---|
| Provides correct solution | 1 |

*Sample Answer:*

$\$17506 \times 108\% = 18906.48$

$\$18906.48 \times 97\% = \$18339.29$

**Question 29 (b)**

| Criteria | Marks |
|---|---|
| Provides correct solution | 1 |

*Sample Answer:*

$\$18339.29 - \$17506 = \$833.29$

**Question 30 (a)**

| Criteria | Marks |
|---|---|
| Provides correct solution | 2 |
| Makes some progress towards the solution | 1 |

*Sample Answer:*

$450 \times 26 = 11700$
$823 \times 12 = 9876$
$Total = 21576$

Taxable income is $21576 - 200 = 21376$

**Question 30 (b)**

| Criteria | Marks |
|---|---|
| Provides correct solution | 1 |

*Sample Answer:*

Medicare levy is $2\% \times 21376 = \$427.52$

**Question 30 (c)**

| Criteria | Marks |
|---|---|
| Provides correct solution | 3 |
| Makes some progress towards the solution | 2 |
| Provides some relevant information | 1 |

*Sample Answer:*

Tax $= (21376 - 18200) \times \dfrac{19}{100} = \$603.44$

Refund is $\$715 - \$603.44 = \$112.56$